

Chapter Four

Stochastic Rainfall Modelling: Poisson-cluster models

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1 Introduction

Since precipitation is the main input to hydrological systems, and rainfall data are not always available at the location of interest, it is useful to have a rainfall model to generate plausible rainfall data that can be used for various hydrological applications.

The physical processes involved in the generation of rainfall are complex, and the mechanisms through which these processes are triggered are various. There are three main such triggers:

- cold and warm fronts (Figure 1);
- convective cells (Figure 2);
- orographic effect or enhancement (Figure 3).

Physical modelling of rainfall therefore depends upon a range of other atmospheric and topographic information defining initial and boundary conditions. It cannot be used to generate long series of rainfall. Stochastic modelling is therefore an obvious tool which has the additional benefit that it allows for the generation of several plausible realisations, and therefore for an estimate of uncertainty.

As we saw in Chapter One, at any scale finer than the month, precipitation time-series are characterised by intermittency.

The presence of periods of 0 rainfall in precipitation (here, we ignore snow) time-series means that linear time-series models are not a particularly appropriate tool to represent the rainfall process at a location over time. This chapter and the next chapter present two popular approaches to the stochastic modelling of rainfall.

2 Motivation for, and approaches to rainfall modelling

What is the purpose of the stochastic modelling of rainfall? Access to rainfall data is important to understand the behaviour of a hydrological catchment behaves. And given that rainfall data



Figure 1: Precipitation from cold and warm fronts

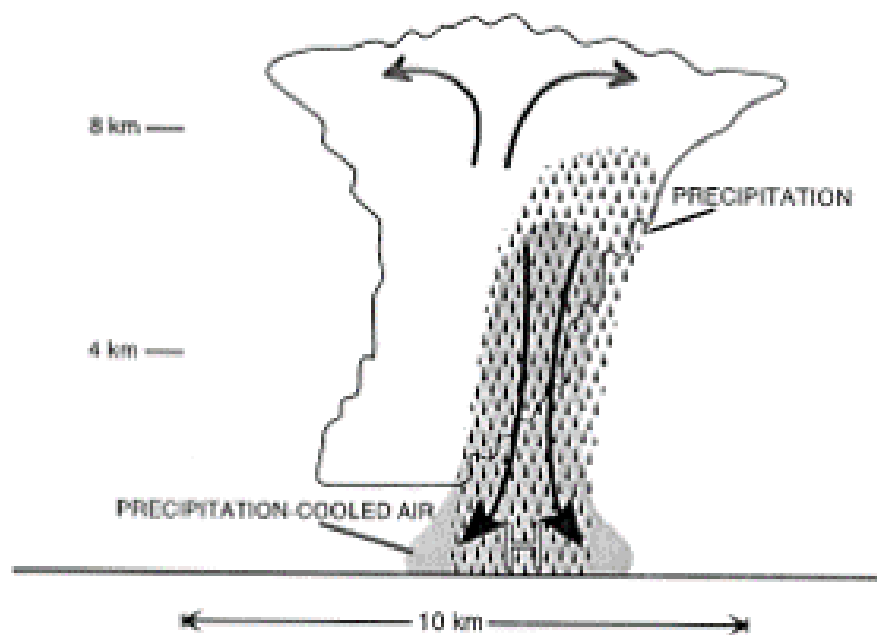


Figure 2: Precipitation from convective activity

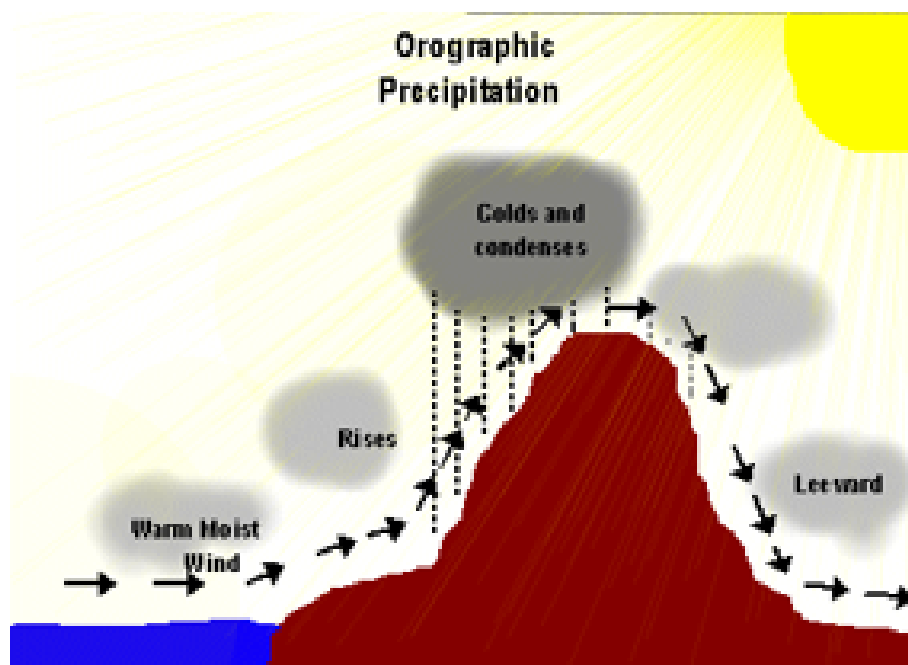


Figure 3: Precipitation generation and enhancement from orography

sets are generally not long enough, or not available at a fine enough scale for the application in question, the possibility of stochastically generating long series of plausible rainfall time-series at scales of interest provides the hydrologist with a very useful tool.

Among the applications for which long time-series and/or fine time-scale data are required are *drought design* and *flood design*. While the first needs only monthly or daily data, the second will at least require hourly, possibly sub-hourly data. In both cases, long time-series are needed for reliable estimates of extremes such as the 1 in 50 year drought or the 1 in 100 year rainfall.

2.1 Design storms for flood design

But now, it will be argued, one could do flood design by using estimates of the extreme rainfalls of interest and inputting these into a catchment model. Since an extreme is only one number, some method will be needed to transform this extreme at the given time-scale (e.g. the concentration time of the catchment of interest) into a sequence of rainfall depths at a time-scale of one hour or less. This is achieved by the *design storm* methodology.

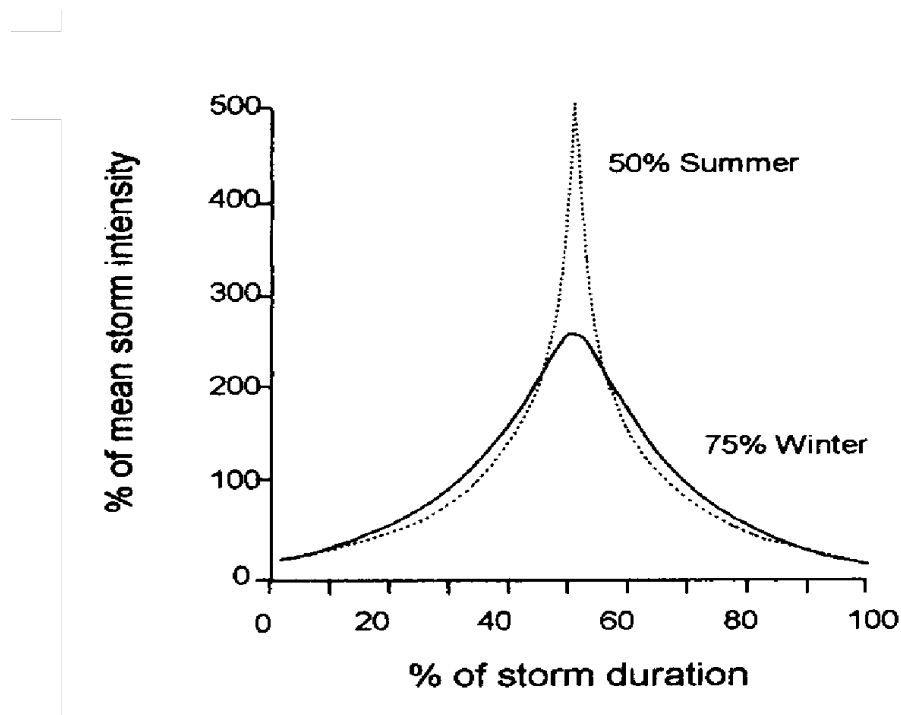


Figure 4: Design Storm with varying peakedness

There are two fundamental problems with this widely-used methodology (?).

- the temporal profile of the design storm (see Figure 4) is unrealistic;
- to obtain a flood of return period T it is not clear what return period should be chosen for the design storm: that it is not identical to the flood return period follows from the role of antecedent conditions.

The second issue is often dealt with by using multiplicative factors that take these antecedent conditions into account, but the uncertainty in this approach is large and not quantified.

2.2 Rainfall modelling for flood design

The use of a stochastic rainfall model addresses both issues. Indeed, the temporal profile of storms is the result of the temporal spread of the contribution of the different cells in the storm. Some cells will overlap, while at other times, there may be short dry periods in the storm. This will lead to multi-modal profiles that are much more realistic than those of a design storm.

In this approach there is no need to make any assumptions about the type of antecedent conditions and their frequency. Indeed, insofar as the model generates time-series with no gaps, the antecedent conditions of an extreme storm will arise from the way the catchment model transforms the previous storms so as to determine the soil moisture conditions prior to this extreme storm.

It is worth noting that, although this chapter focuses upon time-series only, the stochastic rainfall methodology can be extended to produce spatial-temporal rainfall, something that will always be much more realistic than the "blanket rainfall" obtained from a spatial generalisation of a design storm that assumes, unrealistically, that the rainfall intensity is constant over the area of interest.

2.3 Categories of rainfall modelling approaches

There are broadly three categories of stochastic rainfall models. They are either:

- purely statistical models in which, for instance, a standard distribution represents the rainfall depth, with Markov chain occurrences (Kannan and Farook (2015)), or in which a Generalised Linear Model (GLM) relates rainfall to climatological drivers (Chandler and Wheeler (2002)), or in which rainfall is characterised through a marginal distribution and a correlation structure (Papalexiou (2018));
- models in which the *scale-independent* features of rainfall over ranges of temporal scales are modelled explicitly using a small number of parameters, for instance multifractal models (Lovejoy and Schertzer (2013));
- mechanistic models in which the *scale-dependent* features of the rainfall time-series are modelled explicitly (Onof et al. (2000)).

While Chapter Five examines GLMs (first category), this chapter is devoted to the third category. A justification for the approach in this category can be found in looking at the typical spatial structure of rainfall. As shown in Figure 5, small mesoscale structures contain rainfall cells can be identified. They are moreover themselves typically clustered inside large mesoscale structures inside a synoptic area. This spatial structure implies a temporal structuring as such a synoptic area passes over a point (the raingauge). Rainfall cells are typically structures that last less than 30 minutes while small mesoscale areas typically take a few hours to pass over the gauging point.

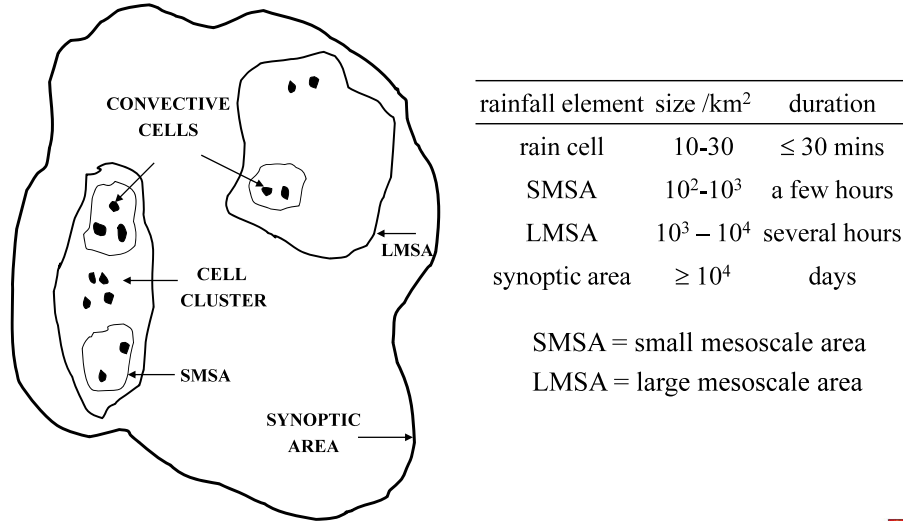


Figure 5: Spatial-temporal structure of rain

Models in this third category typically focus upon representing cells embedded within storms that represent either small or large mesoscale areas. The small contributions to the total rainfall depth made by individual rainfall cells are modelled explicitly. The clustering of cells within storms is then either modelled explicitly (Poisson-cluster models) or arises from a Markov process of cells (Doubly stochastic or Cox models).

The first approach in this third category is more prominent in the literature (since a seminal paper Rodriguez-Iturbe et al. (1987)) and we therefore focus upon it in this chapter. In this chapter, we shall focus upon this approach.

3 Poisson-cluster Rectangular Pulse models

3.1 The clustering mechanism

As explained above, a Poisson-cluster model explicitly represents the clustering of rainfall cells within storms. There are two types of clustering mechanisms (Onof et al. (2000)). These give rise to two alternatives for the continuous-time process representing the underlying precipitation process in continuous time and from which, by aggregation, time-series at any time-scale of interest can be obtained.

These are the Bartlett-Lewis and the Neyman-Scott processes. In both cases, storms arrive according to a Poisson process (hence the generic term of 'Poisson-cluster process'). In the first, the clustering of cells within a storm is represented by generating cells from a second Poisson process that follows the inception of a storm. The time during which cells can arrive (duration of storm activity) is a random variable which limits the size of the storm, i.e. the number of cells it can contain.

In the second, the number of cells in a storm is modelled explicitly as a random variable while the times at which cells arrive are exponentially delayed from the storm start time.

Figure ?? illustrates these two clustering mechanisms.

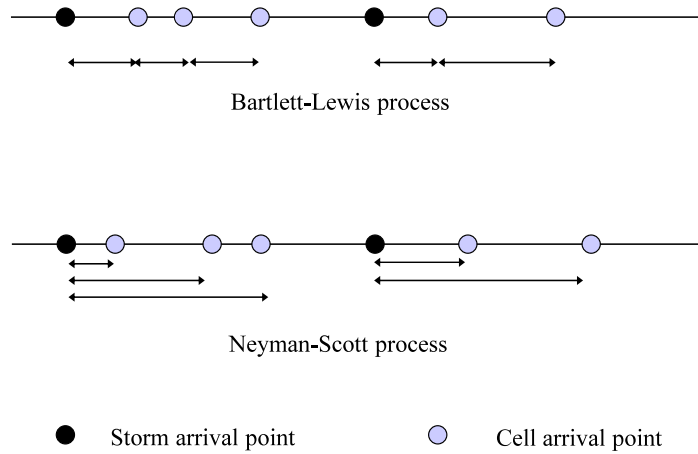


Figure 6: Bartlett-Lewis and Neyman-Scott clustering mechanisms

3.2 The Bartlett-Lewis model

A Poisson-cluster model is then constructed by assigning to each arrival in a Poisson-cluster process a certain amount of rainfall over a certain duration, i.e. what is called a *pulse* of rainfall. This in effect represents the impact of a cell depositing its rainfall as it passes over a raingauge.

Such a pulse is typically chosen as the simplest possible form, namely a *rectangular pulse*. that corresponds to the rainfall rate being constant over the duration of the pulse. Although this might seem unrealistic, it is important to note that the total rainfall is a sum of the contributions from the various pulses which, though overlapping, will enable a noisy hyetograph to be represented.

The overall Bartlett-Lewis model can then be represented as in Figure 7.

This model minimally requires the specification of five random quantities as shown in Figure 8

That is, it requires the specification of:

- the rate at which storms arrive;
- the duration of activity of the storm;
- the rate at which cells arrive during that duration;

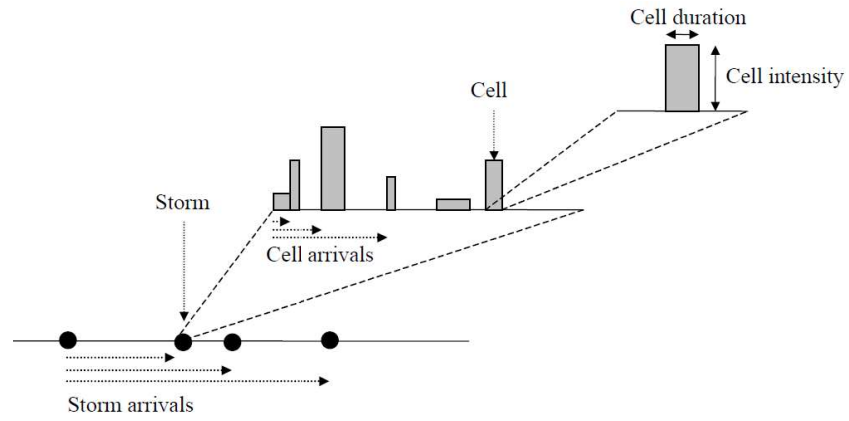


Figure 7: Bartlett-Lewis model

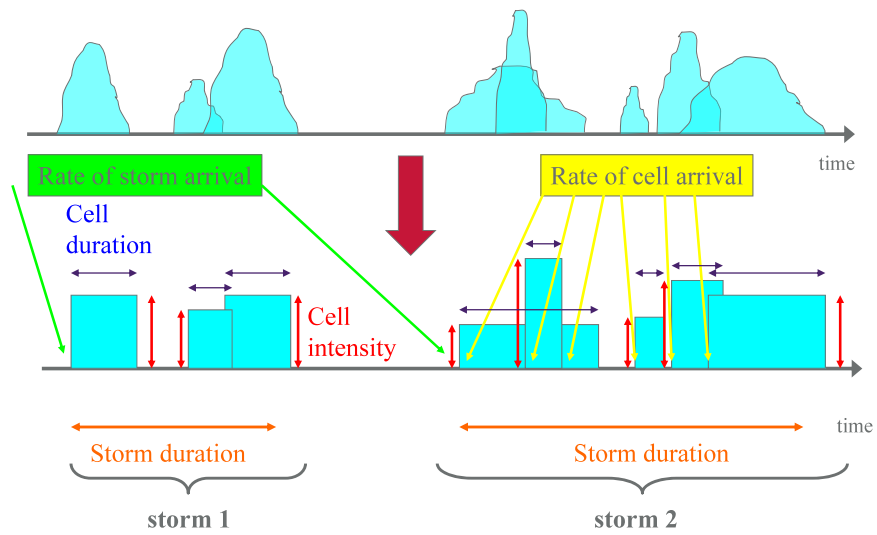


Figure 8: Bartlett-Lewis model structure

- the cell duration;
- the cell intensity.

Minimally, each random quantity can be represented by a random variable or a random process with one parameter only. The basic *Bartlett-Lewis Rectangular Pulse Model* is thus characterised by the five following choices:

- storms arrive according to a Poisson process with parameter λ (hr^{-1});
- storm duration is exponentially distributed with parameter γ (hr^{-1});
- cells arrive according to a truncated Poisson process with parameter β (hr^{-1}) during that duration;
- cell duration is exponentially distributed with parameter η (hr^{-1});
- cell intensity is exponentially distributed with mean μ_X (mm.hr^{-1}).

The last choice can easily be altered so that the cell intensity distribution can be chosen as a two-parameter distribution for instance (gamma, Pareto, Weibull, etc.), in which case a further parameter would be required, e.g. the mean of the square of the rainfall intensity, μ_{X^2} . The other distributions and processes cannot be altered without fundamentally changing the model structure and thereby potentially making the model intractable mathematically. To understand what that means and why it is important, we need to examine how the model is fitted.

4 Model Calibration

A first observation is that, like ARMA models, typical Poisson-cluster models such as the Bartlett-Lewis Rectangular Pulse model are stationary. It follows that they should be fitted to data which do not exhibit obvious trends or seasonality. This problem is typically addressed by fitting the model to individual calendar months, and thereby obtaining 12 sets of parameters for the year. Since this amounts to a model with a substantial number of parameters, i.e. $12 \times 5 = 60\%$ at least, sufficiently long time-series of observations are required. As a rule of thumb, it is generally agreed that 15 to 20 years of data at least are necessary.

Unlike for ARIMA models, it is not possible to obtain the likelihood function in a closed form, for the simple reason that we do not have the distribution of rainfall at a point in a closed form.

However, when sufficiently simple distributions/processes are chosen to represent the random quantities (see above), we obtain mathematically tractable expressions for any time-scale h (in hrs) of the main statistics of the rainfall process, that are of interest in hydro-meteorological applications. That means that we can, in particular, obtain the following expressions, each of which is a function of the :

- mean rainfall depth: $a_1(\lambda, \mu_X, \eta, \beta, \gamma, h)$
- variance of rainfall depths: $a_2(\lambda, \mu_X, \eta, \beta, \gamma, h)$

- auto-correlation of rainfall depths: $a_3(\lambda, \mu_X, \eta, \beta, \gamma, h)$
- proportion of dry intervals: $a_4(\lambda, \mu_X, \eta, \beta, \gamma, h)$
- etc.

The time-scales for which the properties are included are, typically:

- 1, 6, 12 and 24 hours for medium to large catchment hydrological applications;
- 5, 15, 30 minutes and 1, 6, 24 hours for instance, in the case of small and urban catchments

The Generalised Method of Moments will then involve determining the optimal values of parameters $\{\lambda, \mu_X, \eta, \beta, \gamma\}$ whereby the optimality is determined by what minimises an objective function representing the differences between the modelled values of these quantities and the observed values which we denote $o_i(h)$ for $i = 1, \dots$

By allowing different weights ω_i to be assigned to each of the N rainfall properties included in the objective function, we obtain the following expression for the objective function:

$$F(\lambda, \mu_X, \eta, \beta, \gamma) = \sum_{i=1}^{i=N} \omega_i (o_i(h) - a_i(\lambda, \mu_X, \eta, \beta, \gamma, h))^2 \quad (1)$$

Popular choices for the weights are:

- $\omega_i = 1/[o_i(h)]^2$
- $\omega_i = 1/\widehat{Var}(\widehat{O}_i(h))$, i.e. is the inverse of an estimate of the variance of property i

The first choice ensures a standardisation of the additive terms in the objective function so that the scale of each property plays no role.

The second choice has been shown to be theoretically optimal (Jesus and Chandler (2011)). What is needed here is an estimate of the variance of each property. That can readily be obtained by estimating the property for each year of the data set and then using the sample variance of these annual estimates.

Note that some authors (see in Kaczmarek et al. (2014)) have pointed out that it is useful to correct the impact of observations ω_i which are very small in magnitude and poorly reproduced so that $[a_i(\lambda, \mu_X, \eta, \beta, \gamma, h)/o_i(h)]^2$ is large and dominates the objective function. It is then proposed that the objective function reflect this by having additional terms. Although this is not strictly speaking a matter of taking different weights, it can easily be represented by taking, instead of fixed weights ω_i , coefficients $\omega_i(\lambda, \mu_X, \eta, \beta, \gamma, h)$ that depend upon the parameters and the time-scale. Specifically, we could choose:

$$\omega_i(\lambda, \mu_X, \eta, \beta, \gamma, h) = 1/[o_i(h)]^2 + 1/[a_i(\lambda, \mu_X, \eta, \beta, \gamma, h)]^2$$

The equations yielding $a_i(\lambda, \mu_X, \eta, \beta, \gamma, h)$ are non-linear, so the objective function F is thereby highly non-linear. As a consequence, there will typically be a number of local optima. The problem is magnified by the fact that the optimisation takes place in a space of dimension 5 (at least). It is therefore typically the case that the optimisation will be carried out a number of times starting from parameter sets located in different regions of parameter space.

5 The Randomised/Modified models

From the early work on the Bartlett-Lewis Random Parameter Rectangular Pulse Model, it was noted that the reproduction of the proportion of dry intervals was poor over several time-scales. This was put down to the model's insufficient flexibility in the reproduction of intermittency.

To address this problem, it was proposed to include some randomisation so that each storm would in effect have a different temporal scale (Rodriguez-Iturbe et al. (1988)). That is, a storm might typically have more spaced out and longer cells, arriving over a longer period of storm activity.

To bring about that effect, parameter η now becomes a random variable H which is Gamma distributed with parameters α (shape) and $1/\nu$ (scale, in hr^{-1}):

$$H \sim \Gamma(\alpha, 1/\nu) \quad (2)$$

The value of η is fixed throughout a given storm: this defines the temporal scale of the storm. So parameters β and γ also accordingly becomes random variables which are chosen as scaled versions of *eta* with constant scaling factors which become the new parameters. That is, in each storm, a value of β and a value of γ are obtained in terms of two scaling dimensionless parameters κ and ϕ :

$$\beta = \kappa\eta \quad (3)$$

$$\gamma = \phi\eta \quad (4)$$

To summarise therefore, this model has the following 6 parameters defined as follows:

- λ : the rate at which storms arrive;
- μ_X : the mean cell intensity;
- α : shape parameter of the distribution of the cell duration parameter H ;
- $1/\nu$ (hr^{-1}): scale parameter of the distribution of the cell duration parameter H ;
- κ : conditional upon a value $H = \eta$, the Poisson rate of arrival of cells in a storm is $\beta = \kappa\eta$;
- ϕ : conditional upon a value $H = \eta$, the exponential parameter of the duration of storm activity is $\gamma = \phi\eta$;

This is generally referred to as the *Random Parameter Rectangular Pulse Bartlett-Lewis Model*, or the *Modified Rectangular Pulse Bartlett-Lewis Model*. Note that a more recent randomised/modified model has shown to lead to improvements in the reproduction of rainfall statistics (Kaczmariska et al. (2014), Onof and Wang (2020)), particularly for extreme values. The difference is that, like η, β and γ, μ_X now becomes a random variable. so that the parameters are now:

- λ : the rate at which storms arrive;
- α : shape parameter of the distribution of the cell duration parameter H ;
- $1/\nu$ (hr^{-1}): scale parameter of the distribution of the cell duration parameter H ;
- κ : conditional upon a value $H = \eta$, the Poisson rate of arrival of cells in a storm is $\beta = \kappa\eta$;
- ϕ : conditional upon a value $H = \eta$, the exponential parameter of the duration of storm activity is $\gamma = \phi\eta$;
- ι : conditional upon value $H = \eta$, the mean cell intensity is $\mu_X = \iota\eta$;

This choice means that cells which are longer will typically have smaller intensities, which reflects a well-known observational fact about rain cells. This model is sometimes referred to as the *Random μ_X Rectangular Pulse Bartlett-Lewis Model*.

6 Model Validation

6.1 Parameter Identifiability

Although no likelihood function is available to provide confidence intervals for the parameters, an approximate theory was formulated to make up for this (Wheater and S. (2006)). As a result, it is possible to plot profiles of the objective function as function of any of the model parameters. As Figure 9 shows, the profiles of some parameters such as λ indicate that it is well-identified because there is a well-defined and not too long interval of values of the parameter that are below the 95% threshold, i.e. that are such that, with 95%, the optimal parameter lies in that segment (and similarly, of course with a somewhat longer interval for the 99% confidence level).

This is not the case for several other parameters, e.g. parameters α and ν of this Random Parameter Bartlett-Lewis Rectangular Pulse Model. In fact, it turns out that, while the ratio of these parameters is fairly well-defined, they are not individually well-identified. This problem is sometimes addressed by putting an upper limit on the value of either of these parameters, Note that it is not the case that their ratio is actually constant, so we cannot dispense with either parameter.

We also see some large confidence intervals for other parameters. This illustrates the difficulty in satisfactorily calibrating these Poisson-cluster models.

The problem of parameter identifiability can also be inferred indirectly by looking at the optimal parameters for each month (see table 1 in Figure 10) and examining whether their variability is relatively smooth or not.

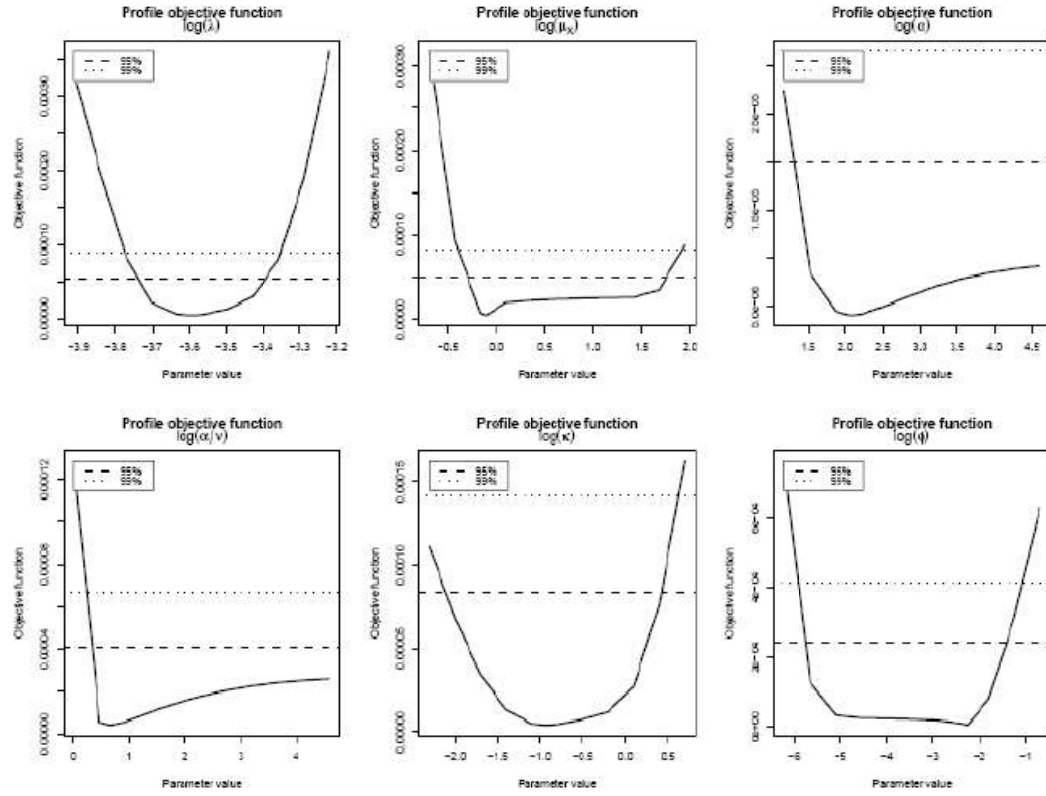


Figure 9: Parameter Identification for the random parameter model

Table 1
Parameters of the randomised model for individual months (R2)

Month	λ (h ⁻¹)	κ	ϕ	α	ν (h)	$E(\eta)$ (h ⁻¹)	μ_x (mm h ⁻¹)	μ_c	$E(\beta)$ (h ⁻¹)	$E(\gamma)$ (h ⁻¹)
January	0.0296	0.6467	0.1266	3.1986	1.2780	2.5027	0.7383	6.1067	1.6186	0.3169
February	0.0248	0.8832	0.0941	4.0378	1.1276	3.5808	0.7346	10.3849	3.1625	0.3370
March	0.0259	0.7761	0.0759	2.9840	0.6543	4.5602	0.7235	11.2320	3.5393	0.3459
April	0.0234	0.6018	0.0876	3.8451	1.0225	3.7603	0.9593	7.8700	2.2630	0.3294
May	0.0242	0.2599	0.0231	2.7570	1.1658	16.6264	2.7414	12.2452	4.3212	0.3843
June	0.0211	0.2932	0.0385	2.5250	0.2792	9.0432	2.2768	8.6245	2.6512	0.3477
July	0.0111	0.0421	0.0064	2.2894	0.2476	9.2473	4.1869	7.5515	0.3889	0.0594
August	0.0117	0.0448	0.0081	2.8811	0.6885	4.1846	3.4405	6.5560	0.1875	0.0337
September	0.0228	0.4186	0.0706	2.4801	0.4369	5.6771	1.7325	6.9327	2.3763	0.4005
October	0.0216	0.3570	0.0758	2.6583	0.7611	3.4926	1.2922	5.7096	1.2470	0.2648
November	0.0303	0.6135	0.1414	3.2886	1.3079	2.5145	0.9966	5.3395	1.5426	0.3555
December	0.0318	0.8103	0.0874	2.6078	0.4631	5.6311	0.8632	10.2678	4.5628	0.4923

Figure 10: Optimal Parameters for the Random Parameter model at Elmdon airport (Birmingham)

6.2 Reproduction of Standard Statistics

The most important tool in assessing the goodness-of-fit of such a stochastic model is to examine the performance of the model in reproducing standard statistics $a_i(\dots, h)$ of the model for time-steps of interest. Some of these statistics were used in the calibration and so we would expect them to be well reproduced. Others are not generally as well reproduced. Figure 11 shows model performance in terms of two statistics which were not used in the fitting. Here, the analytical (model) estimates of the statistics are shown with $\pm 5\%$ confidence intervals which is a shortcut for a proper representation of parameter uncertainty.

We can see that for most months both statistics are well reproduced, except for the 12-hour variance in June and July. When statistics are poorly reproduced, this is always a motivation for revisiting:

1. the calibration method, or
2. the model specification.

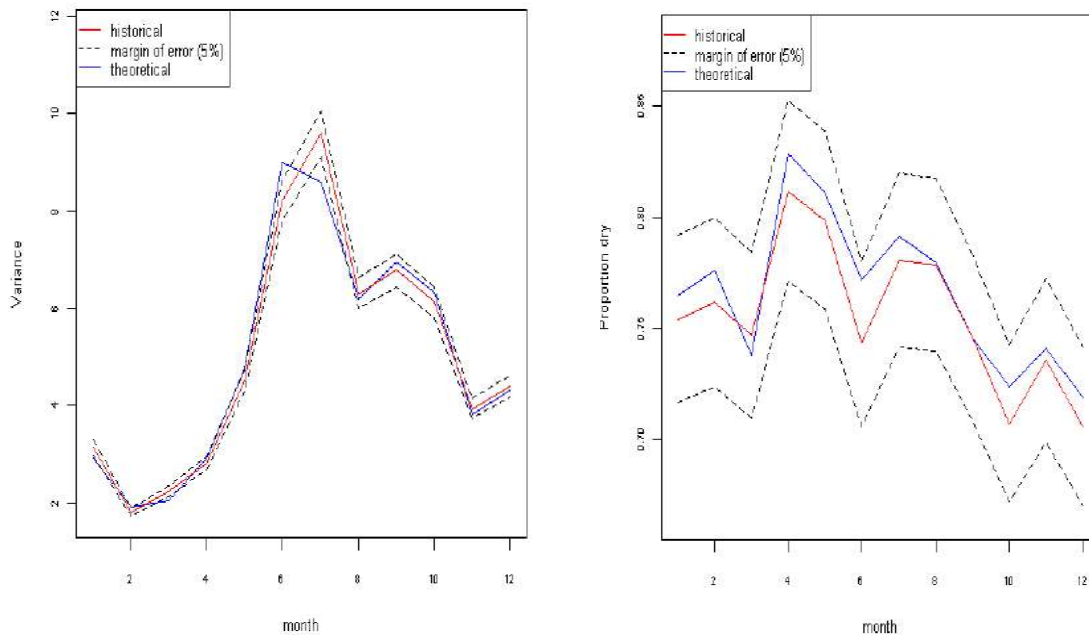


Figure 11: Standard Statistics at Holbaek (Denmark): Variance and Proportion Dry at $h=12$ hrs

For instance, research upon the mathematical expressions of the rainfall statistics, as well as on the empirical estimates of these statistics has enabled the improvement of the performance of the Random μ_X Rectangular Pulse Bartlett-Lewis Model as shown in Figure 12. This is a case where the calibration method was improved by showing that a broader parameter space (specifically, a broader range of possible values of parameter α) could be considered, and by improving the estimates of the observational statistics.

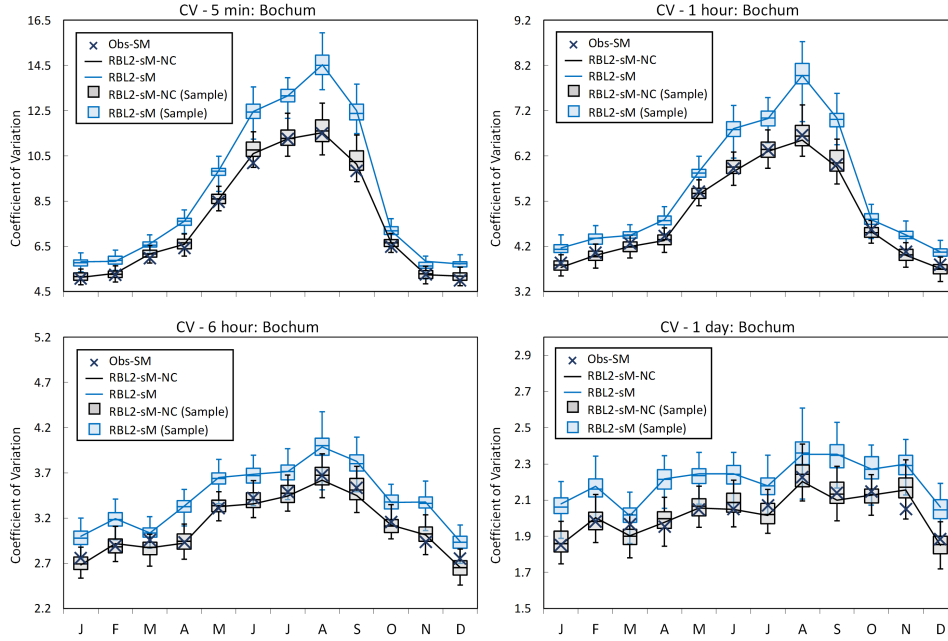


Figure 12: Coefficient of variation (CV) by month at Bochum (Germany): the observed vs. the fitted estimates using the original and new versions of the Random μ_x model, respectively in blue and black

If you compare Figures 11 and 12, you will see two different kinds of representations of the modelled statistics and their uncertainty.

In the former case, we saw that the *analytical* estimates of the statistics (obtained from model equations) were plotted together with estimates of the effect of *parameter uncertainty* upon these statistics. In the latter case, the *simulated* values of the statistics are shown with box-and-whisker plots. This does not reflect parameter uncertainty, but *stochastic uncertainty*.

The difference between these two types of uncertainty is the following:

- *Parameter uncertainty* tells us about how much confidence we have in the way the model is calibrated.
- *Stochastic uncertainty* is due to the limited length of the simulated time-series. It is useful in practice to enable the impact of the limited length of records to be taken into account.

The second uncertainty will get smaller as the length of the time-series increases. For an infinitely long time-series, the *simulated* model statistics would be equal to the *analytical* ones. The first uncertainty can only be reduced by improving model calibration.

6.3 Reproduction of extreme-value statistics

Since one of the most important type of applications of a stochastic rainfall model is in hydrological design, it is important to validate its performance in terms of its ability to reproduce what is known about the extreme values of the observed rainfall depths at relevant scales. These

scales range from *5mins* to a few hours for urban hydrological applications and from *1hour* to 24 hours for hydrological design in medium to large catchments. The return periods of relevance are respectively of the order of 20 to 100 years in the first case and 100 to 200 years in the second.

Reproduction of extremes is however not at all guaranteed by the way Poisson-cluster stochastic models are fitted. That is because these models are fitted using standard statistics which describe the mean behaviour of the rainfall process: all these statistics are expected values of random variables, depending either upon the rainfall depth distribution or the distribution of duration of wet or dry spells. As such, they represent the mean behaviour of these random variables and therefore are unlikely to be closely related to the extremes of the rainfall depth distribution.

It is therefore not surprising to find that, for a collection of near-optimal parameters sets in terms of the objective functions defined earlier, there can be a wide-range in the extreme values that are generated, as shown in Figure 13.

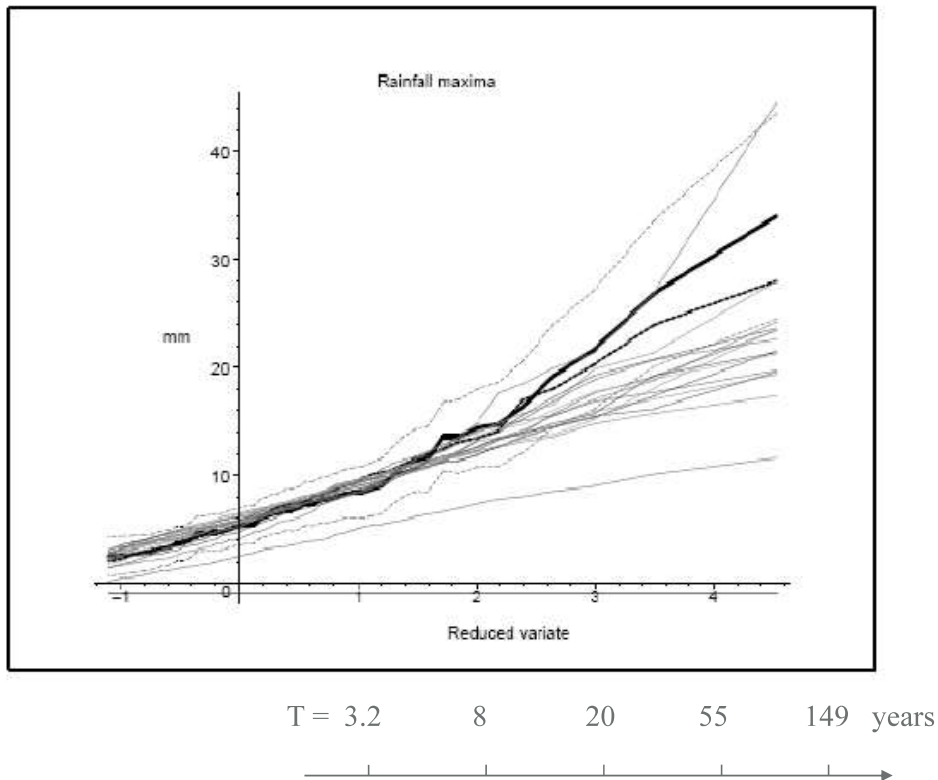


Figure 13: Gumbel plot of the hourly extreme rainfall depths for a range of near-optimal parameter sets. The thickest line represents the observations; the other lines are each obtained by simulation from a parameter set with the best singled out as thicker

The question is: how does one identify, among near-optimal parameter sets, that which produces the best extreme-value reproduction? As Figure 13 shows, this can be done by selecting among near-optimal parameter sets that which produces the closest extremes to those found in the observational data set. This however requires that a large number of simulations be carried out with randomly selected near-optimal parameter sets. Note that, while this should lead to a good reproduction of extremes at a given time-scale, the question of how well the extremes are reproduced at other time-scales is an open one.

Alternatively, or additionally, it is possible to seek an improvement of extreme-value performance by including the third-order moment or the coefficient of skewness of rainfall depths at several relevant scales into the objective function. As Figure 14 shows, by including coefficients of skewness, an impressive extreme-value performance can be obtained at all scales. REVERSE FIGURE CAPTION

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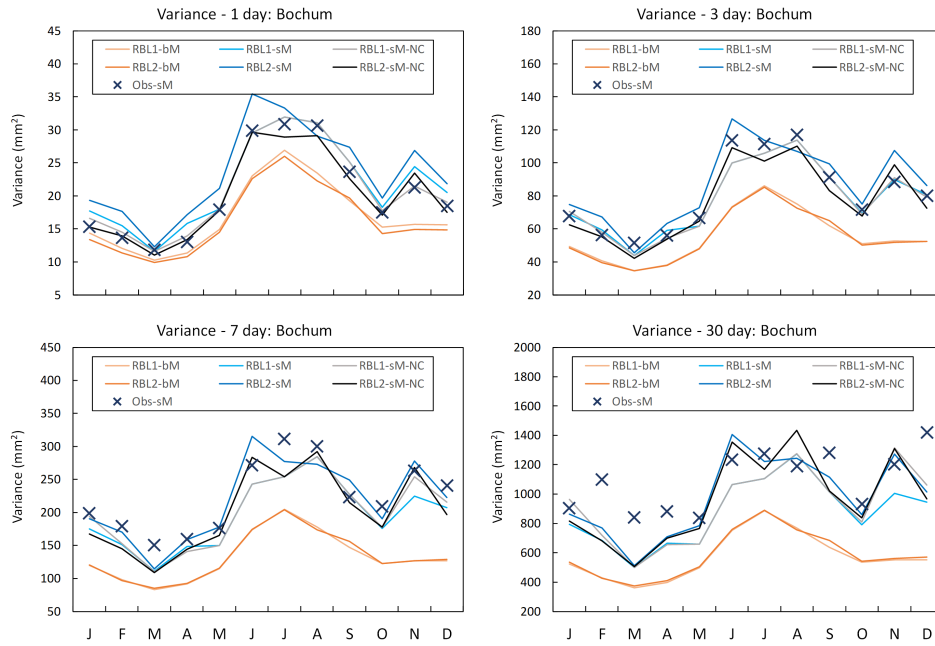


Figure 14: Observed (round markers) and simulated (lines) return levels of rainfall at multiple time-scales at Bochum. The simulated estimates are from two versions (old in blue; new in black) of the Random μ_X model fitted with skewnesses included in the objective function. The median, 95 and 5 percentile return levels obtained from 250 simulations, each of 69 years, are plotted with solid and dashed lines, respectively.